

# All Palindromes

Time limit: 1.00 s

Memory limit: 512 MB

Given a string, calculate for each position the length of the longest palindrome that ends at that position.

## Input

The only line contains a string of length  $n$ . Each character is one of a–z.

## Output

Print  $n$  numbers: the length of each palindrome.

## Constraints

- $1 \leq n \leq 2 \cdot 10^5$

## Example

| Input       | Output                |
|-------------|-----------------------|
| ababbababaa | 1 1 3 3 2 4 6 8 5 5 2 |